LectureNotes Offline

Gujarati / Hindi lecture notes on Snapdragon, with no internet needed

Snapdragon® AI Lab Build & Present Challenge 2026 (Qualcomm) | Target: Snapdragon-powered HP AI PC

Summary

An offline tool that turns Gujarati/Hindi lecture recordings into a transcript, revision notes and flashcards, all on the student's own laptop.

The problem

Many engineering students in India attend lectures that mix Gujarati, Hindi and English technical terms. Taking notes while listening is hard, and most AI note tools need a steady internet connection, upload recordings to the cloud, and work poorly in regional languages.

The solution

A student loads a lecture recording and the app runs everything locally:

1	**Transcribe**	Speech to text in Gujarati, Hindi or English using Whisper.
2	**Summarize**	Short revision notes using a small on-device LLM (Llama 3.2 3B).
3	**Flashcards**	Question and answer cards for quick revision.
4	**Export**	Transcript, Markdown notes and a flashcards CSV.

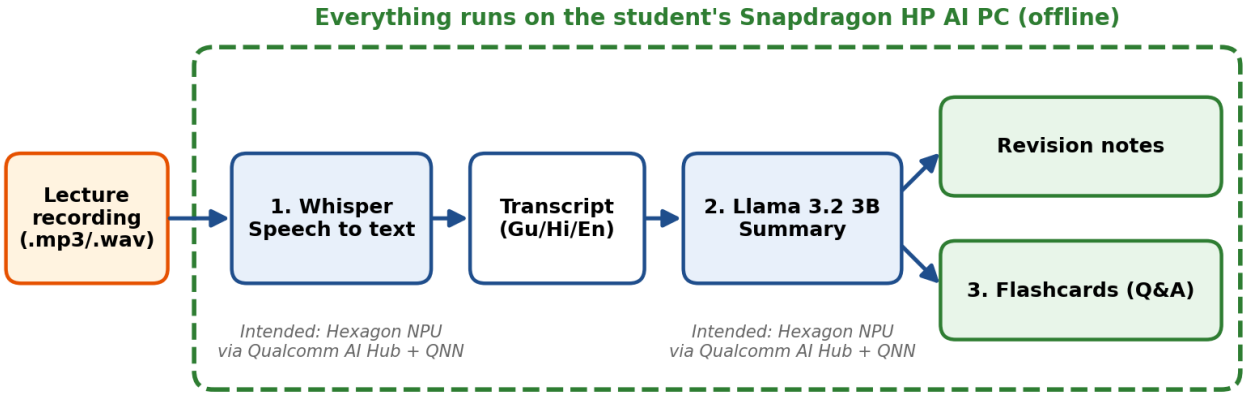


Figure 1: How a lecture recording becomes notes and flashcards.

Why on-device / Snapdragon

- **Works offline:** no dependence on campus Wi-Fi or mobile data.
- **Private:** recordings, transcripts and notes never leave the device.
- **Low power:** the Hexagon NPU is designed to run these models efficiently, which matters for battery life during a full day of lectures.
- **Ready path:** Whisper and Llama 3.2 3B are available on Qualcomm AI Hub, so deployment is compiling them for the NPU and running through ONNX Runtime with the QNN Execution Provider.

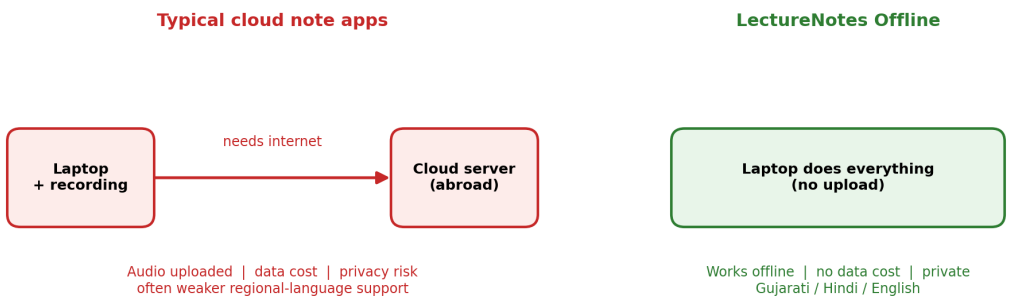


Figure 2: Cloud note apps vs. LectureNotes Offline.

Tech stack

Whisper, Llama 3.2 3B Instruct, Qualcomm AI Hub, ONNX Runtime (QNN), Python.

What the output looks like

Example output (illustrative, in English) - lecture: PN junction diode

NOTES

- A diode conducts easily in forward bias and blocks current in reverse bias.
- A rectifier uses diodes to convert AC into pulsating DC.
- Key terms: forward bias, reverse bias, rectifier

FLASHCARDS

Q: What does a diode do in reverse bias? A: It blocks current.	Q: What is a rectifier used for? A: Converting AC to DC.

Figure 3: Illustrative example only (not real measured output).

Current status

The prototype runs the full pipeline (transcribe, summarize, flashcards) on CPU. The NPU path is the intended deployment. **[EDIT before submitting: state whether you tested on a Snapdragon device, and any measured results.]**

Limitations

- Gujarati transcription is usually less accurate than Hindi or English, especially with technical terms and mixed-language speech.
- A 3B-parameter model can write weaker summaries in Gujarati; review notes and flashcards before studying from them.
- Whisper can make mistakes on names, formulas and equations.

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